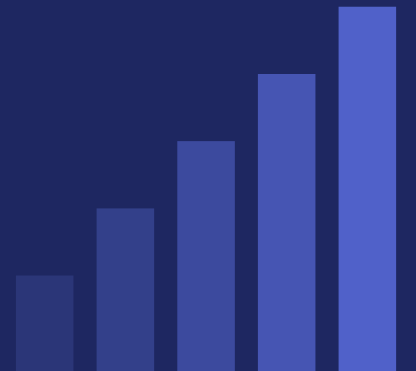


GCI WORLD 2026 • PROOF OF CONCEPT

Proactive Customer Retention

An ML-Based Strategy for Company A

Mitigating Telecom Churn Through Device Upgrades and Plan Optimization



Presented by: Associate Consultant

Executive Summary

● THE CHALLENGE

Company A suffers a critical churn rate of nearly 50% — 49,562 churners out of 100,000 customer records.

● THE DISCOVERY

Churn correlates strongly with two bottlenecks: handset obsolescence (equipment age) and a sharp decline in monthly usage minutes.

● THE SOLUTION

An automated XGBoost model identifies high-risk customers at 60.5% test accuracy and 0.65+ ROC-AUC, enabling targeted proactive retention.

● FINANCIAL IMPACT

A projected net annual revenue lift of \$1,520,000 USD through a targeted retention upgrade framework.

Exploratory Data Analysis — The Churn Crisis

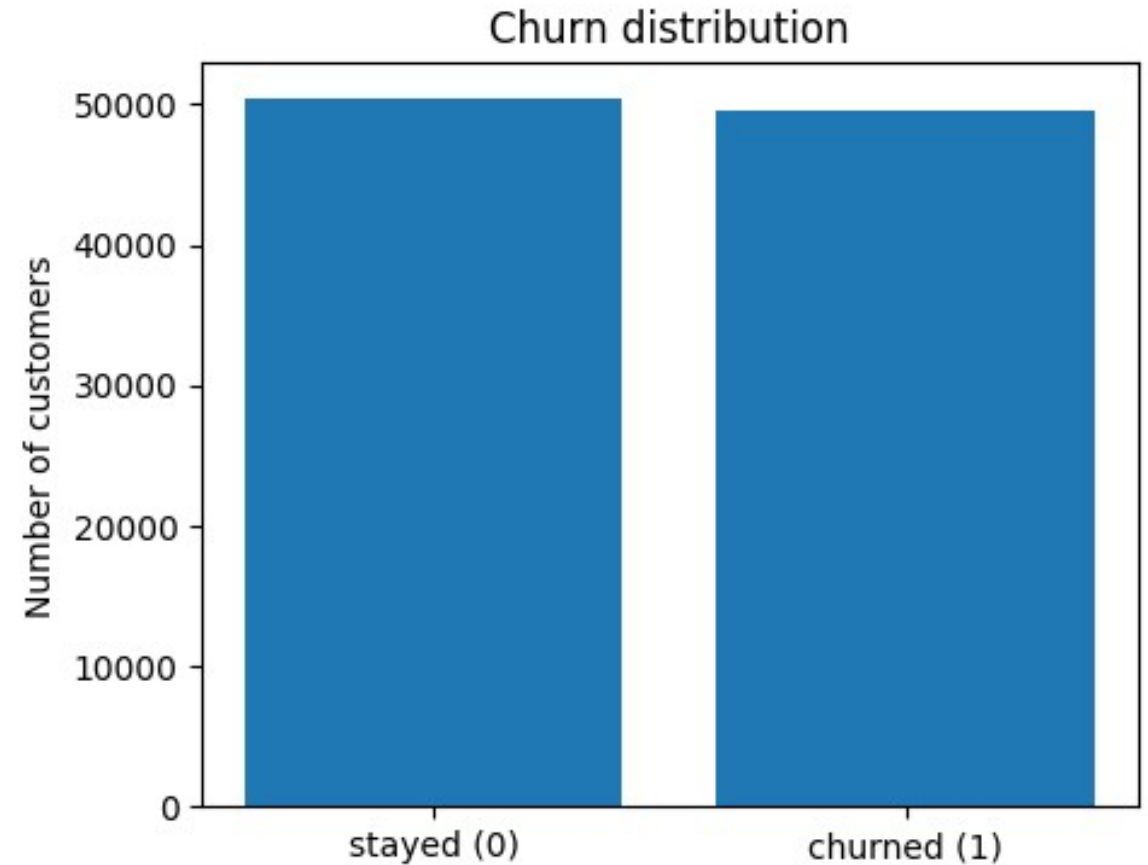
100,000

Total customer records analyzed

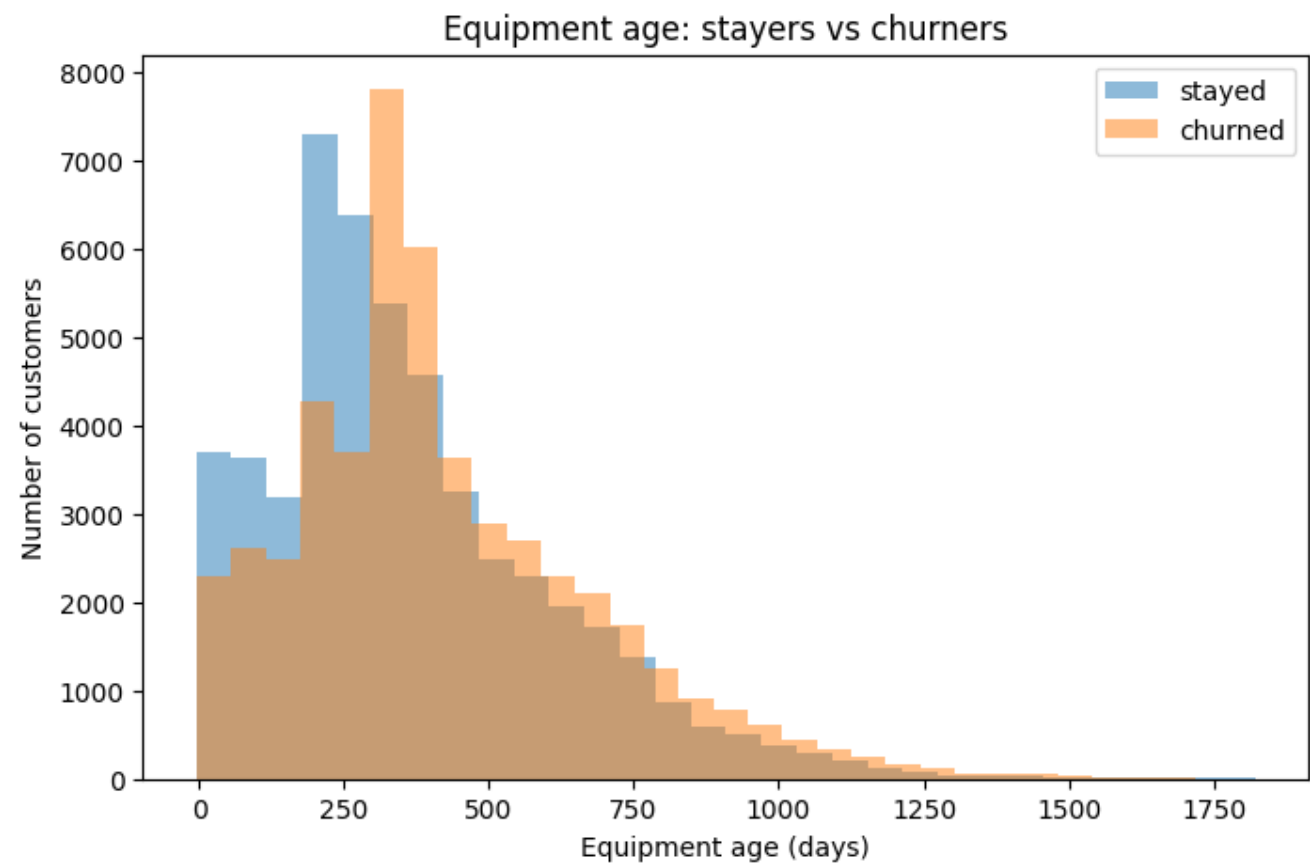
- **50,438** Active customers (50.4%)
- **49,562** Churned customers (49.6%)

BUSINESS IMPACT

Losing half of the acquired customer base creates a massive revenue leak and drives up Customer Acquisition Costs (CAC) — confirming that a systematic retention framework is non-negotiable.



Deep Dive — Handset Obsolescence (eqpdays)



DATA FINDINGS

363.3 days average equipment age

Active customers

421.1 days average equipment age

Churned customers

STRATEGIC INTERPRETATION

Customers holding onto devices longer than 1.1 years exhibit significantly higher churn risk — indicating device degradation, lack of 5G/modern features, or end-of-contract vulnerability where competitors offer free device switches.

Deep Dive — Behavioral Usage Drop (change_mou)

Customers do not suddenly leave — they gradually disengage from the service weeks before officially churning.

ACTIVE CUSTOMERS

+6.1 min

average increase in monthly usage

CHURNED CUSTOMERS

−151.7 min

catastrophic average decline in monthly usage

STRATEGIC INTERPRETATION

A contraction in minutes of use (change_mou) serves as an immediate behavioral indicator of customer disengagement — and an early-warning signal for intervention.

Predictive Churn Engine (XGBoost)

Objective: predict the probability of a customer churning next month, based on 10 core billing and behavioral features.

FEATURES USED

change_mou

mou_Mean

eqpdays

rev_Mean

months

change_rev

totmrc_Mean

drop_vce_Mean

hnd_price

custcare_Mean

MODEL PERFORMANCE

60.5%

Baseline Test Accuracy

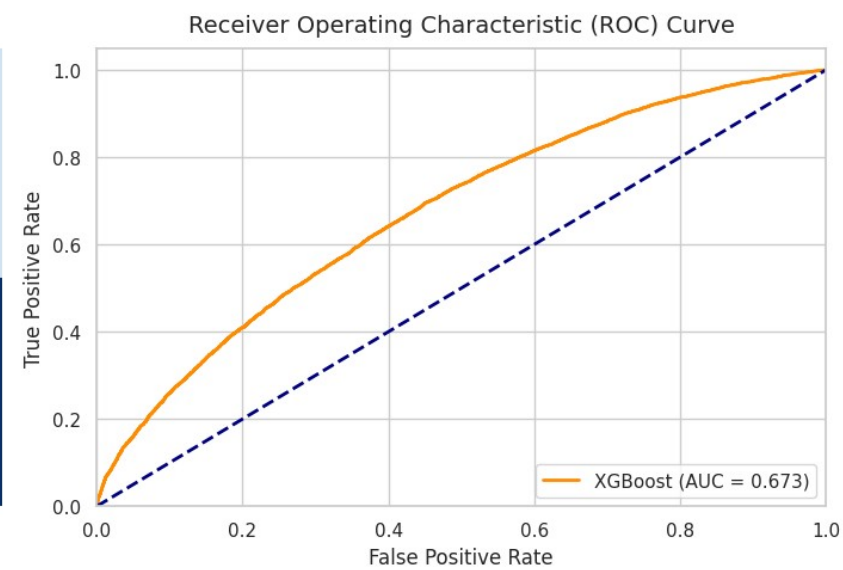
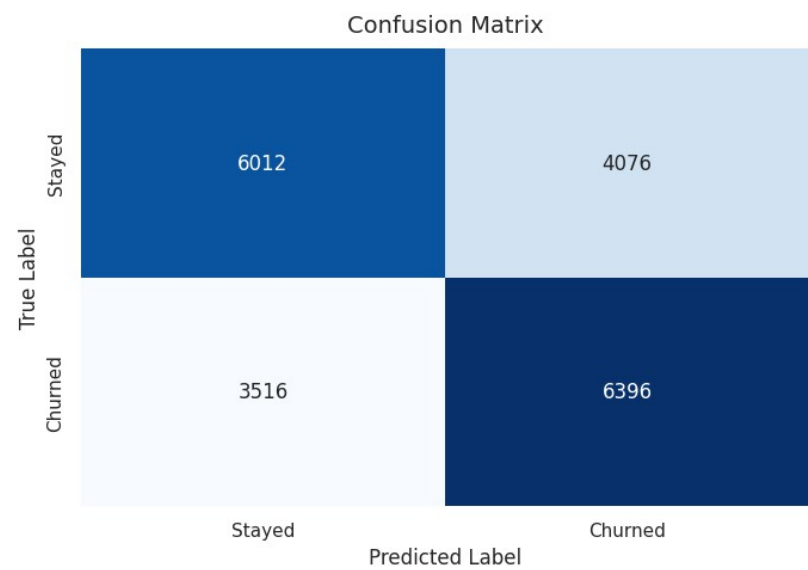
of churn predictions correctly classified

~0.655

ROC-AUC Score

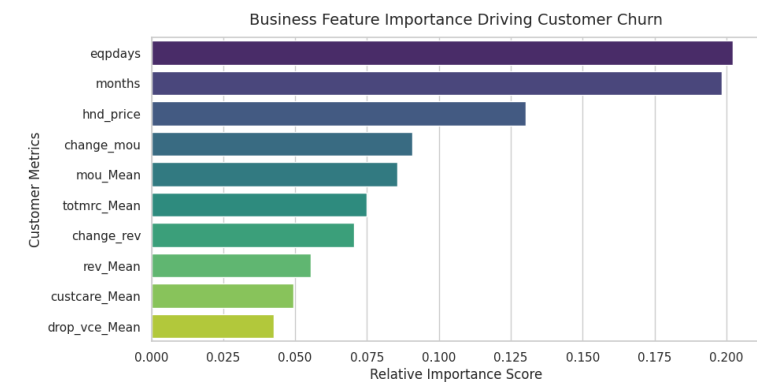
reliable statistical power, well above random guessing

Model Verification & Feature Importance



TOP CHURN DRIVERS IDENTIFIED BY MODEL

- 1 **change_mou** — Usage Drop Velocity — primary behavioral trigger
- 2 **mou_Mean** — Overall Usage Level — low-volume users are highly volatile
- 3 **eqpdays** — Device Age — primary physical/infrastructure trigger



The Business Proposal

Proactive Device Upgrade & Plan Optimization

CORE STRATEGY

Instead of standard reactive win-back campaigns — expensive and inefficient — Company A will deploy an automated "Proactive Upgrade & Plan Right-Sizing" engine.

HOW IT WORKS

1

Weekly Scoring

The XGBoost model scores the entire customer base every week.

2

Risk Flagging

Customers meeting the dual threshold — Equipment Age > 400 days AND Change in MOU < -100 min — are automatically enrolled in a targeted retention cohort.

3

Offer Intervention

Each flagged customer receives a personalized \$50 USD handset upgrade discount voucher, coupled with an optimized data/voice package adjustment.

Quantified Business Impact & ROI

Financial returns quantified, based on the 100,000-customer pool

Financial Element	Value Formula / Calculation	Impact Amount
Target Population	Top 20,000 highest-risk customers identified by ML	20,000 customers
Historical Churn Risk	70% baseline churn within this high-risk cluster if untreated	14,000 lost customers
Retention Success Rate	Conservative 30% saved via device upgrade voucher	4,200 customers saved
Preserved Annual Revenue	4,200 saved × \$50 monthly revenue (rev_Mean) × 12 months	+\$2,520,000 USD
Retention Campaign Cost	20,000 targeted customers × \$50 voucher incentive cost	-\$1,000,000 USD
Net Financial Lift	Preserved Revenue (\$2.52M) – Campaign Cost (\$1.00M)	+\$1,520,000 USD

SYSTEM IMPLEMENTATION

Implementation & Consulting Fees

To transition this Proof of Concept into full production, we propose the following enterprise fee structure:

ONE-TIME FEE

\$150,000 USD

AI Core Engine Implementation

Includes data pipeline integration, model deployment on Company A infrastructure, and custom threshold alerting setups.

MONTHLY RETAINER

\$10,000 / mo

Consulting & Optimization

Includes monthly model retraining for seasonal trends, feature drift tracking, and continuous campaign performance optimization.

VALUE PROPOSITION

Total Year-One Investment: \$150K setup + \$120K annual support = **\$270,000** — a minor fraction (~17.7%) of the \$1.52M net savings, paying for itself within the first 3 months of operations.